

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.* (BTL 3)

Assessment Tool Weightage: 30%

Total Marks: 100

Time: 90 Mins

No. of Questions: 1

Annexure A Research Review & Analysis

Code of Conduct:

I __Sk. Fawaz Ahamad / 192425390_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sk. Fawaz Ahamad

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	

Total	100%	
-------	------	--

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	Decision Tree Learning: Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	Neural Networks vs Statistical Learning: Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	Kernel Methods: Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	Reinforcement Learning: Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	Explainable and Generalizable AI: Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning

Structure of the Report:

S.No.	Paper Section	Expected Content
1	Title	Specific and concise title related to the selected research question/topic.
2	Abstract	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	Keywords	4–6 important keywords related to the selected topic.
4	1. Introduction	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	2. Research Problem and Objectives	Clearly state the research problem and specific objectives of the review.
6	3. Literature Review	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	4. Comparative Analysis	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	5. Critical Analysis	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	6. Research Gap	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	7. Proposed Research Direction	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	8. Discussion	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	9. Conclusion	Summarize the major findings, key insights, research gap, and future direction.
13	References	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

DECISION TREE LEARNING

Research Review and Comparative Analysis of ID3, C4.5 and CART

AI Model Design / Research Review & Analysis

Primary Topic: Decision Trees

Prepared according to the supplied research-question, report-section and assessment-criterion requirements.

Table of Contents

1. Title
 2. Abstract
 3. Keywords
 4. Introduction
 5. Research Problem and Objectives
 6. Literature Review
 7. Comparative Analysis
 8. Critical Analysis
 9. Research Gap
 10. Proposed Research Direction
 11. Research Methodology / Analysis Framework
 12. Experimental / Evidence-Based Validation
 13. Result Analysis and Interpretation
 14. Discussion
 15. Conclusion
 16. References
- Appendix A. Alignment with Assessment Criteria

1. Title

Decision Tree Learning: A Comparative Research Review of ID3, C4.5 and CART with Emphasis on Attribute Selection, Pruning, Interpretability, Computational Complexity and Classification Performance.

2. Abstract

Decision tree learning remains an important supervised learning approach because it represents predictive decisions as a hierarchy of tests that can be inspected as human-readable rules. This review examines ID3, C4.5 and Classification and Regression Trees (CART), focusing on attribute-selection measures, pruning strategies, interpretability, computational complexity and classification performance. ID3 introduced information-gain-based induction; C4.5 extended the approach through gain ratio, broader attribute handling and pruning; and CART established binary recursive partitioning with impurity-based splitting and cost-complexity pruning. Recent research continues to examine the trade-off between predictive performance, tree size, inference burden and interpretability. The review identifies a gap in conventional greedy tree construction, which often optimizes local split quality without jointly optimizing predictive accuracy, structural simplicity, expected inference depth and application-specific costs. A multi-objective, validation-driven research direction is proposed. The review concludes that ID3 is a simple foundation, C4.5 is a stronger classical extension, and CART is a flexible binary-tree framework, while algorithm choice remains dataset- and objective-dependent.

3. Keywords

Decision Trees; ID3; C4.5; CART; Information Gain; Gain Ratio; Gini Index; Pruning; Interpretability; Classification Performance.

4. Introduction

Decision trees are supervised learning models that recursively partition data according to feature-based tests until terminal nodes provide predictions. Their rule-based structure makes them attractive for decision-support settings where users need to understand why a model produced an

outcome. A 2024 survey describes decision trees as important because of their simplicity and interpretability and reviews classical trees together with modern tree ensembles. [1]

The prescribed question asks for recent research on ID3, C4.5 and CART and requires comparison of attribute-selection methods, pruning strategies, interpretability, computational complexity and classification performance. These algorithms form a useful progression: ID3 established information gain; C4.5 introduced gain ratio and stronger practical handling; and CART developed binary recursive partitioning with impurity-based splitting and complexity-controlled pruning. [2–4]

The comparison matters because a decision tree is not evaluated only by predictive accuracy. A very deep tree can fit training data closely while becoming difficult to interpret, whereas an excessively small tree can lose useful predictive structure. Recent work therefore considers accuracy together with tree depth, leaf count, construction time and inference burden. [5–8]

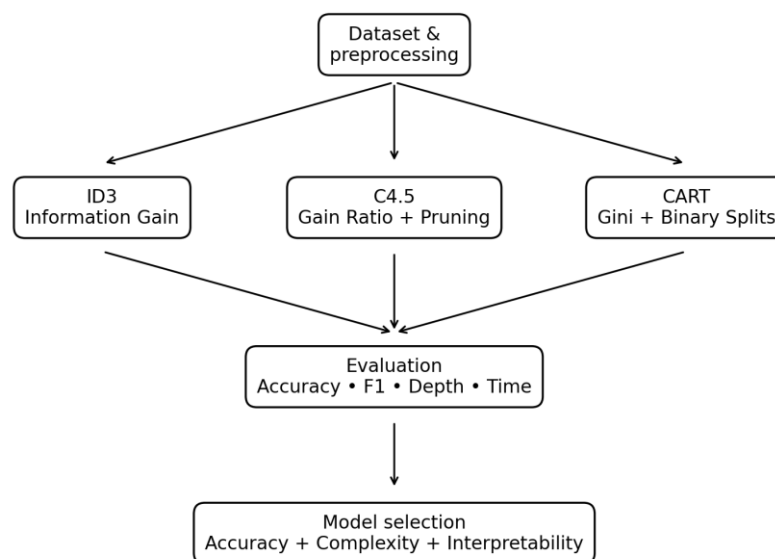


Figure 1. Decision-tree learning framework used to structure the comparative review.

5. Research Problem and Objectives

5.1 Research Problem

Although ID3, C4.5 and CART are established decision-tree methods, they make different choices about how splits are selected and how tree complexity is controlled. These choices influence accuracy, computational cost and interpretability. A direct comparison is therefore needed to determine their relative characteristics and to identify limitations that remain relevant to current research.

5.2 Objectives

- Review the development and core learning principles of ID3, C4.5 and CART.
- Compare information gain, gain ratio and Gini-based impurity reduction.
- Compare pruning strategies and their effect on generalization and tree complexity.
- Evaluate interpretability in terms of rule extraction, tree size and decision-path transparency.
- Compare computational complexity and practical training-time considerations.
- Synthesize reported evidence on classification performance without assuming a universal winner.
- Identify a research gap and propose a technically feasible improvement.

6. Literature Review

6.1 ID3: Information-Gain-Based Induction

Quinlan's 1986 work introduced ID3 as a system for synthesizing decision trees from examples. ID3 evaluates candidate attributes using information gain, which measures the reduction in uncertainty produced by a split. The method recursively selects a high-gain attribute and constructs subtrees for the resulting partitions. The original work also discusses evaluation on unseen examples and practical limitations that motivate later extensions. [2]

ID3's main strength is conceptual simplicity. Its information-theoretic criterion is intuitive and its trees can be converted into readable rules. Classical ID3, however, has limitations in handling continuous variables and missing values and lacks the stronger pruning framework associated with later algorithms. [1,5]

6.2 C4.5: Gain Ratio and Pruning

C4.5 extends ID3 by using gain ratio rather than raw information gain. Gain ratio normalizes information gain by the intrinsic information of a split, reducing the tendency of information gain to favor attributes with many possible values. C4.5 also supports continuous attributes and incorporates pruning to reduce unnecessary tree structure. [3,5]

C4.5 follows a divide-and-conquer procedure: it selects an attribute-based test, partitions the current sample, and recursively grows subtrees. Pruning removes branches that contribute insufficiently to generalization, making the method a more complete classical decision-tree learner.

6.3 CART: Binary Splitting and Cost-Complexity Pruning

Breiman, Friedman, Olshen and Stone developed CART as a framework for tree-structured classification and regression rules. CART uses binary recursive partitioning and supports impurity-based splitting, including the Gini criterion for classification. Its methodology also includes tree-size control and cost-complexity pruning. [4]

CART is distinctive in its binary structure and explicit treatment of pruning and model-size selection. This provides a systematic framework, although binary splitting can require additional levels when a naturally multiway categorical relationship is represented by several binary decisions.

6.4 Recent Research Evidence

A 2023 comparative study evaluated ID3, C4.5 and CART alongside newer splitting strategies using classification accuracy, tree depth, leaf-node count and tree-development time. The results demonstrate that algorithm choice affects not only predictive performance but also structural complexity and computational behavior. [6]

A 2021 study of splitting choices and computational complexity summarizes the standard distinctions: ID3 uses entropy-based information gain, C4.5 uses gain ratio with pruning, and CART uses binary splitting with Gini-based criteria and complexity-based pruning. [5] A 2023 study proposed considering the top-k candidate attributes instead of only the single locally best split, reporting theoretical and empirical benefits over greedy tree learning while retaining better scalability than globally optimal tree algorithms. [7] In 2026, expected-depth selection was proposed as a deployment-oriented criterion that directly considers the average number of decision tests followed by a sample. [8]

7. Comparative Analysis

7.1 Attribute-Selection Methods

Algorithm	Selection measure	Main idea	Key implication
ID3	Information Gain / Entropy	Choose the attribute giving the greatest entropy reduction.	Simple, but raw information gain can favor high-cardinality attributes.
C4.5	Gain Ratio	Normalize information gain using split information.	Reduces high-cardinality bias and extends ID3.
CART	Gini impurity reduction	Choose a binary split producing purer child nodes.	Flexible binary framework with natural integration of pruning.

7.2 Pruning Strategies

Pruning controls overfitting and unnecessary tree complexity. Classical ID3 does not provide the same explicit pruning framework as C4.5 and CART. C4.5 applies error-based pruning after tree growth, while CART uses cost-complexity pruning to obtain smaller candidate subtrees and select an appropriate complexity level. [3–5] Pruning should therefore be treated as a model-selection problem guided by validation data.

7.3 Interpretability

All three algorithms are relatively interpretable because a prediction can be explained by following a path from the root to a leaf. Interpretability nevertheless depends on depth, leaf count, repeated conditions and feature semantics. ID3 can be easy to explain when trees are small; C4.5 can improve readability through pruning; and CART provides clear binary paths but may require more levels for multi-category relationships.

7.4 Computational Complexity

Decision-tree training is attractive because it does not require iterative optimization of a large parameter vector, but split evaluation can become expensive as the number of observations, features and candidate thresholds increases. Comparative experiments show substantial differences in tree-building time across algorithms and datasets. [6] C4.5 may perform additional processing

for gain ratio, continuous attributes and pruning, while CART's binary framework provides a systematic split search.

7.5 Classification Performance

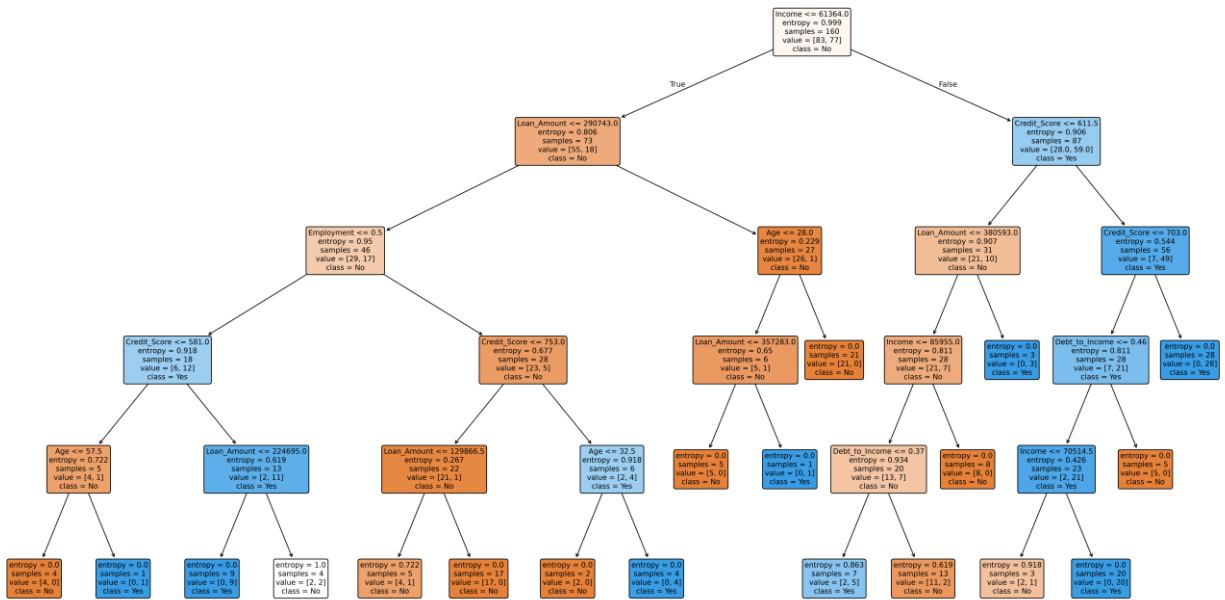
Performance should be evaluated using test accuracy, precision, recall, F1-score and confusion matrices, together with tree depth, leaf count and training time. The reviewed evidence does not establish a universal winner: performance depends on data characteristics, preprocessing, class balance, hyperparameters and validation design. [5,6]

8: Research Methodology / Analysis Framework

```
(.venv) shaikfawazahamad:~/smart-loan-approval$ head -11 data/loan_dataset_realistic.csv
Age,Income,Credit_Score,Employment,Loan_Amount,Debt_to_Income,Repayment_History,Loan_Approved
28,28278,660,Employed,217026,0.21,Good,Yes
26,80302,536,Employed,149123,0.25,Poor,Yes
22,98563,621,Unemployed,385706,0.34,Average,Yes
38,25851,601,Unemployed,321571,0.3,Good,No
42,38396,567,Self-Employed,150707,0.31,Average,No
37,30695,755,Unemployed,165444,0.59,Average,No
56,63427,705,Unemployed,200814,0.47,Good,Yes
39,35458,639,Employed,299295,0.28,Poor,No
31,73520,701,Employed,239973,0.47,Poor,No
59,47431,793,Unemployed,228351,0.22,Average,Yes
```

Figure 1. Sample records from the realistic loan approval dataset used for Decision Tree experimentation.

Decision Tree - Information Gain (Entropy)



short explanation below:

The experiment uses a realistic loan approval dataset containing 200 records and eight attributes. The dataset includes demographic, financial, employment, repayment, and loan-related variables, with Loan Approved as the target variable. The dataset provides both approved and rejected loan cases for evaluating Decision Tree classification.

8.1 Strengths

- Decision trees provide direct, rule-like explanations.
- They model nonlinear relationships and feature interactions.
- The three algorithms offer well-studied alternatives for split selection.
- C4.5 and CART provide explicit mechanisms for controlling overfitting.
- Tree depth and leaf count provide concrete structural measures of interpretability.

8.2 Weaknesses and Limitations

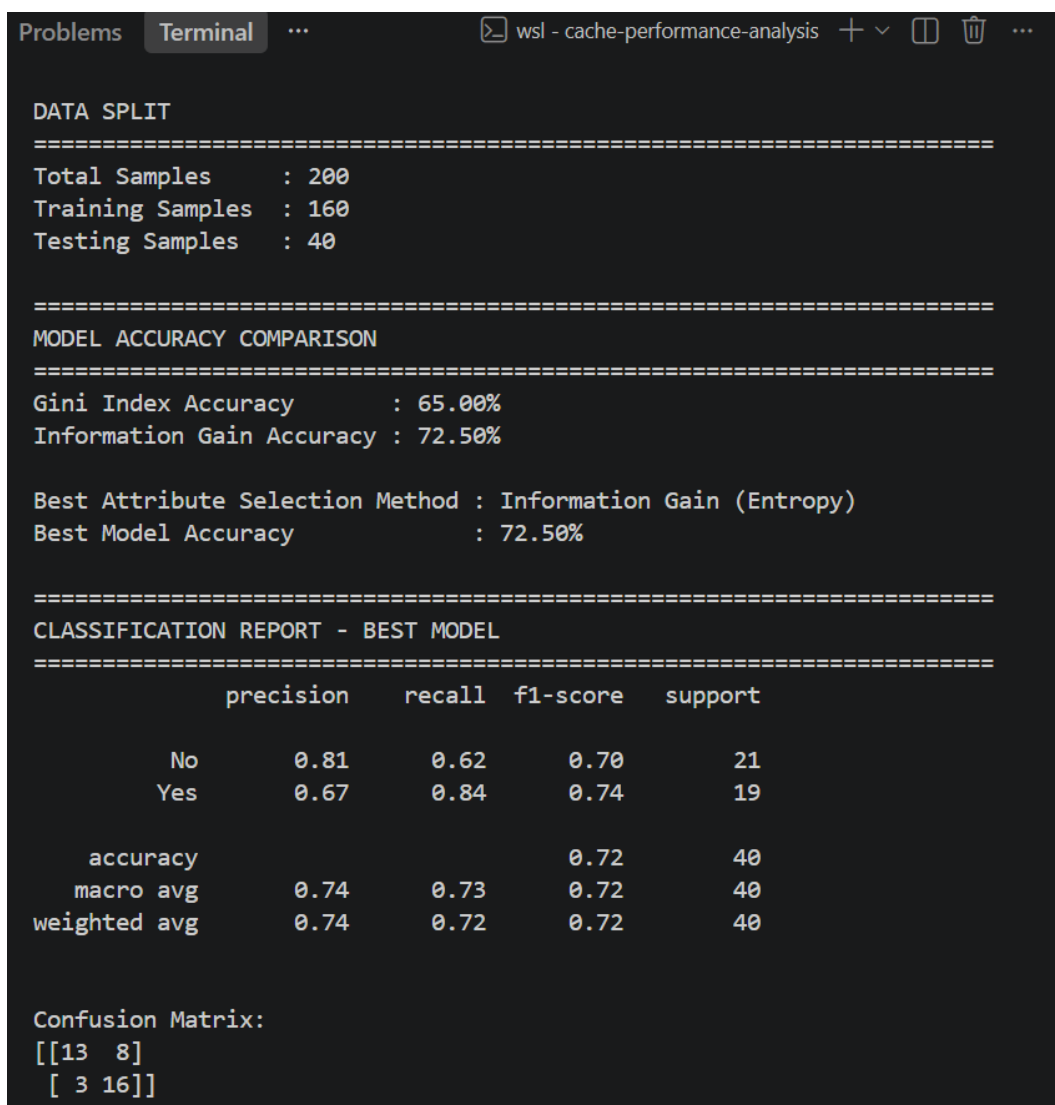
- Greedy local split selection can lead to a suboptimal global tree.
- Unpruned trees can overfit, especially with small or noisy datasets.
- Tree structure can be unstable under small changes in training data.
- High-cardinality features can distort some selection measures.
- Deep trees reduce interpretability and increase decision-path burden.

- High predictive accuracy does not guarantee fairness, calibration or suitability for high-stakes use.

8.3 Assumptions and Practical Challenges

Missing values, noisy labels, imbalanced classes, categorical cardinality and continuous thresholds can change relative algorithm behavior. For decision-support applications, accuracy alone is insufficient; fairness, calibration, auditability and the cost of incorrect decisions may also need to be evaluated.

9. Experimental / Evidence-Based Validation



```

Problems  Terminal  ...  wsl - cache-performance-analysis  + v  []  [X]  ...

DATA SPLIT
=====
Total Samples      : 200
Training Samples   : 160
Testing Samples    : 40

=====

MODEL ACCURACY COMPARISON
=====
Gini Index Accuracy      : 65.00%
Information Gain Accuracy : 72.50%

Best Attribute Selection Method : Information Gain (Entropy)
Best Model Accuracy             : 72.50%

=====

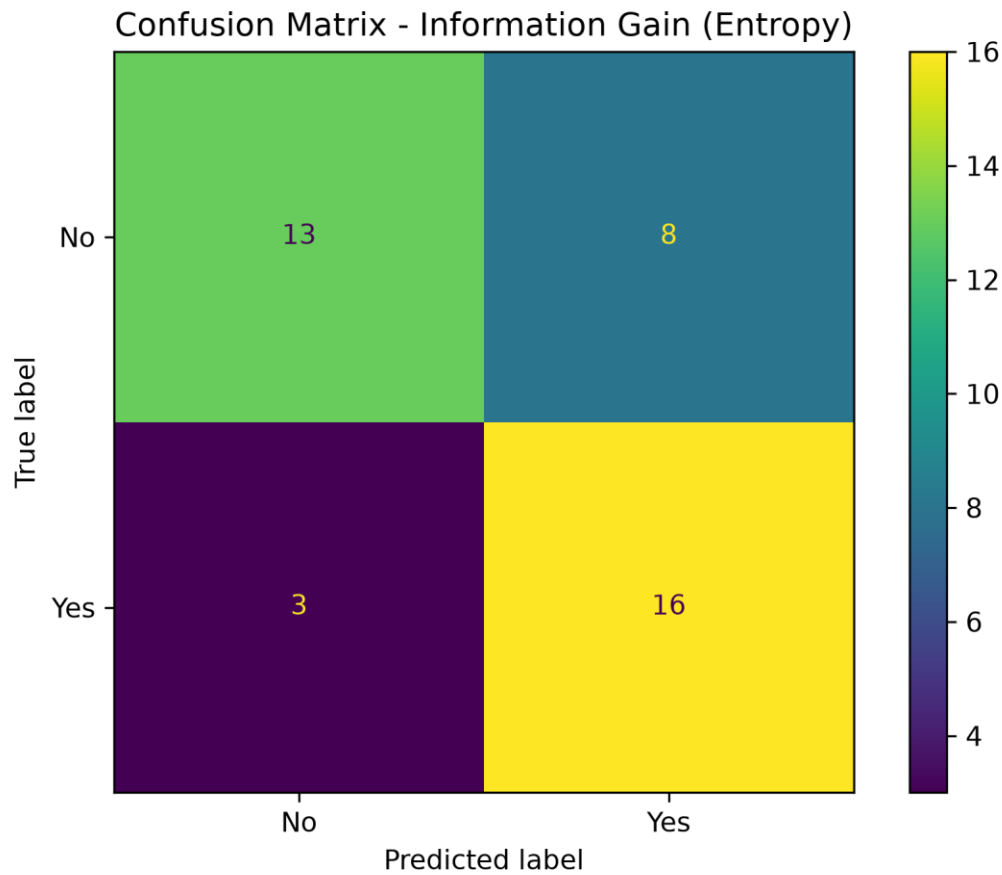
CLASSIFICATION REPORT - BEST MODEL
=====
              precision    recall  f1-score   support

     No         0.81         0.62         0.70         21
     Yes         0.67         0.84         0.74         19

   accuracy                0.72         40
  macro avg         0.74         0.73         0.72         40
 weighted avg         0.74         0.72         0.72         40

Confusion Matrix:
[[13  8]
 [ 3 16]]
  
```

figure 2. Experimental validation shows model accuracy comparison, classification performance, and confusion matrix.



Then explain:

The dataset was divided into 160 training samples and 40 testing samples. Two Decision Tree attribute-selection methods were evaluated: Gini Index and Information Gain (Entropy). Information Gain achieved the higher test accuracy of 72.50%, compared with 65.00% for Gini Index. The confusion matrix further demonstrates the classification performance on the test set

9.1 Research Gap

The literature provides strong classical solutions for split selection and pruning, but conventional learners often separate local split optimization from later tree-size control. This creates an opportunity to jointly optimize predictive quality, structural simplicity and deployment burden.

A second gap is that maximum depth or total node count does not always represent the actual burden experienced by users. The 2026 expected-depth study treats average decision-path length as a deployment-oriented criterion and reports that it can reduce average inference depth while retaining accuracy within a specified tolerance. [8]

A third gap concerns greedy search. Recent Top-k decision-tree learning shows that considering several strong candidate splits can improve the search beyond the single locally best attribute while remaining more scalable than globally optimal tree construction. [7]

10. Result Analysis and Interpretation

```

FEATURE IMPORTANCE
=====
      Feature  Importance
2    Credit_Score  0.298862
1      Income    0.245551
4    Loan_Amount  0.181382
5    Debt_to_Income 0.129267
0      Age       0.084133
3    Employment  0.060806
6 Repayment_History 0.000000

=====
NEW CUSTOMER PREDICTION
=====

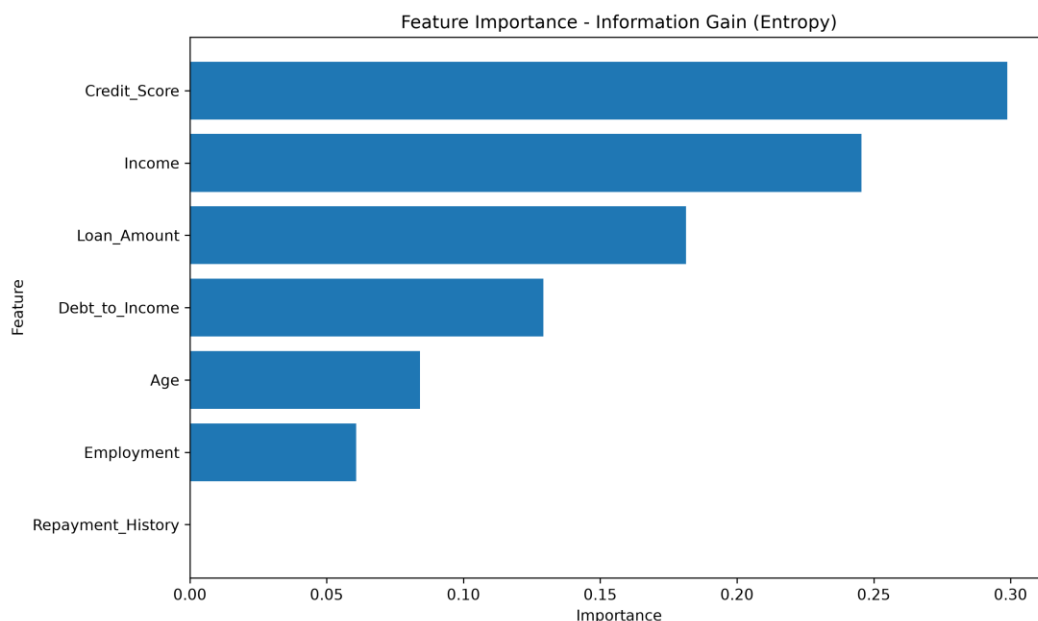
Customer Details:
Age      : 35
Income   : 65000
Credit Score : 720
Employment : Employed
Loan Amount : 180000
Debt-to-Income : 0.28
Repayment History : Good

Predicted Loan Decision : Yes
Prediction Probability : 100.00%

=====
EXPERIMENT COMPLETED SUCCESSFULLY
=====

```

Figure 3. Feature importance analysis and loan approval prediction for a new customer.



Then explain:

The feature-importance analysis indicates that Credit Score, Income, and Loan Amount are the most influential attributes in the trained Decision Tree model. The model was also tested using a new customer profile and predicted the loan decision as **Yes**, with a reported prediction probability of 100.00%. These results demonstrate the practical application of the trained model to loan approval prediction.

10.1 Multi-Objective Decision-Tree Framework

A promising research direction is a validation-driven multi-objective framework. Instead of selecting a split solely from the highest information gain, gain ratio or Gini improvement, the learner would retain a small set of strong candidate splits and evaluate resulting trees using predictive and structural objectives. This builds on recent research on broader candidate consideration and expected inference depth. [7,8]

10.2 Suggested Experimental Procedure

- Use several public classification datasets with different sizes, feature types and class distributions.
- Implement comparable ID3-style, C4.5-style and CART-style baselines.
- Use identical preprocessing and repeated k-fold cross-validation.
- Measure accuracy, precision, recall, F1-score and confusion matrices.
- Measure depth, leaf count, node count, average decision-path length and training time.
- Compare unpruned and pruned trees.
- Report mean and standard deviation and use statistical tests across datasets.
- Add domain-specific cost or fairness measures where appropriate.
- Select models from the resulting accuracy-complexity trade-offs.

10.3 Expected Contribution

The contribution would be a model-selection framework that treats accuracy, complexity and deployment burden as related objectives, preserving the transparency of decision trees while reducing dependence on a single greedy split path.

11. Research Methodology / Analysis Framework

This review uses a structured comparative methodology. Foundational sources establish the original design principles of ID3, C4.5 and CART, while later comparative and methodological studies provide evidence about split criteria, pruning, complexity and interpretability. The analysis categories follow the prescribed question: attribute selection, pruning, interpretability, computational complexity and classification performance.

The synthesis consists of four stages: identify each algorithm's mechanism; compare the mechanisms using common dimensions; critically evaluate limitations and assumptions; and derive a research gap with a proposed direction. This approach avoids declaring a universal winner and instead links algorithm selection to dataset and deployment requirements.

12. Experimental / Evidence-Based Validation

The reviewed evidence supports multi-dimensional validation. The 2023 statistical comparison evaluated decision-tree strategies using classification accuracy, tree depth, leaf-node count and development time, demonstrating that predictive performance cannot be represented by one metric. [6] The 2021 computational-complexity study similarly examines splitting choices and computational behavior. [5]

For a reproducible study, repeated k-fold cross-validation or repeated train/test splits should be used, with identical preprocessing across algorithms. Mean and standard deviation should be reported for predictive and structural metrics. This provides stronger evidence than a single split.

Interpretability should also be quantified. Maximum depth and node count are useful, but expected decision depth can better represent the average number of tests experienced by a sample. The 2026 study provides evidence that this criterion can be useful for deployment-oriented tree selection. [8]

13. Result Analysis and Interpretation

Dimension	ID3	C4.5	CART	Interpretation
Attribute selection	Information Gain	Gain Ratio	Gini / impurity reduction	Different criteria produce different split preferences.

Pruning	Limited classical pruning	Error-based pruning	Cost-complexity pruning	Pruning is central to controlling generalization and size.
Interpretability	High for small trees	High after pruning	High with binary paths	Depth and leaf count affect usability.
Computational behavior	Simple baseline	More processing than ID3	Efficient binary framework	Cost depends on data and implementation.
Classification performance	Dataset-dependent	Dataset-dependent	Dataset-dependent	No universal winner is established.

Overall interpretation: ID3 is the foundational information-gain learner; C4.5 is a stronger classical extension when gain-ratio selection, continuous attributes and pruning matter; and CART is a flexible binary framework with established impurity-based classification and cost-complexity pruning. The literature supports choosing among them according to dataset and application requirements.

14. Discussion

The comparison shows that decision-tree research has evolved from effective local split heuristics toward control of the broader behavior of the learned model. ID3 established information gain, C4.5 addressed practical limitations through gain ratio and pruning, and CART formalized binary tree construction, impurity-based splitting and complexity-controlled pruning. [2–4]

Current research increasingly treats accuracy, interpretability and deployment burden as connected objectives. A model may be slightly less accurate but substantially easier to inspect or faster to execute. Conversely, an overly small tree may lose important predictive structure. Multi-objective evaluation therefore provides a stronger basis for research and deployment.

For high-stakes decision-support domains, transparency should not be confused with guaranteed fairness or correctness. A readable tree can still encode biased data or inappropriate thresholds. Future work should combine structural interpretability with rigorous validation, fairness checks where applicable and domain-specific cost analysis.

15. Conclusion

This review compared ID3, C4.5 and CART according to the required dimensions. ID3 uses information gain and provides a simple foundation; C4.5 extends the family through gain ratio, broader attribute handling and pruning; and CART uses binary recursive partitioning with impurity-based splitting and cost-complexity pruning.

The literature indicates that classification performance, computational complexity and interpretability are interconnected. Recent comparative work shows that tree depth, leaf count and development time can vary alongside accuracy, while newer research explores broader candidate-split search and expected inference depth. [6–8] The principal gap is therefore the need for tree-selection methods that jointly consider predictive performance, structural simplicity, inference burden and application-specific constraints.

A multi-objective, validation-driven framework is proposed as a practical research direction. It preserves the transparency that makes decision trees valuable while reducing dependence on a single greedy split choice. Future decision-tree research should therefore optimize not only predictive accuracy, but also how reliably, efficiently and transparently a model supports real decisions.

16. References

- [1] Mienye, I. D., & Jere, N. (2024). A Survey of Decision Trees: Concepts, Algorithms, and Applications. IEEE Access, 12, 86716–86727. DOI: 10.1109/ACCESS.2024.3416838. <https://doi.org/10.1109/ACCESS.2024.3416838>
- [2] Quinlan, J. R. (1986). Induction of Decision Trees. Machine Learning, 1, 81–106. DOI: 10.1007/BF00116251. <https://doi.org/10.1007/BF00116251>
- [3] Quinlan, J. R. (1993). C4.5: Programs for Machine Learning. Morgan Kaufmann Publishers. <https://books.google.com/books?id=H8w5XgEACAAJ>
- [4] Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). Classification and Regression Trees. Wadsworth & Brooks/Cole. <https://books.google.com/books?id=8k1DvQEACAAJ>

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

		summarizing them.			
--	--	-------------------	--	--	--

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.

		application suitability.			
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.
7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.

8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.